

Measuring Teacher Attrition with Labour Force Survey Data

The Case of the United States

José Elías Durán Roa

Data Cluster · Directorate for Education and Skills



Why this project

- Which of the factors we can observe correlate most strongly with the decision to leave teaching?
- Re-estimate the teacher leaving rate against the published figures of the official surveys.

Contribution

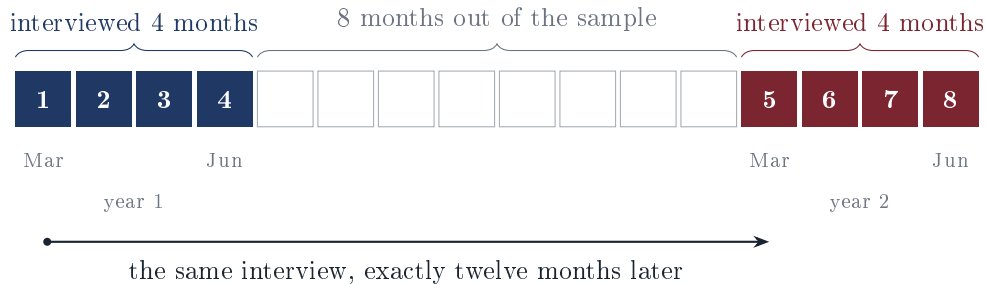
1. **A new series.** The longest **continuous** series of teacher attrition to reach the present: one estimate for **every year from 1997 to 2024**, constructed from twenty-eight consecutive supplements of the Current Population Survey.
2. **A measurement discussion.** A systematic treatment of the ways a longitudinal household survey identifies leaving: the linking of individuals across interviews, the alternative definitions of a leaver, and the assumptions each definition entails.
3. **Substantive evidence.** A characterization of who leaves teaching and of the observed factors most strongly associated with leaving, together with the destinations of leavers and the heterogeneity of attrition across states.

The CPS replicates the official picture — and does it every year

	CPS	NTPS (official)	Difference
Measurement	every year, 1997–2024	occasional waves	
Example wave: 2020–21, the latest NTPS			
Female (%)	75.5	76.8	−1.3
Master’s degree or higher (%)	57.0	61.0	−4.0***
Average age (years)	44.0	42.9	+1.1***
Teachers observed in the wave	3,855	39,633	

Public school teachers. CPS: supplements of 2021–2022 (school years 2020–21 and 2021–22), weighted. NTPS: National Teacher and Principal Survey 2020–21 (NCES 2022-113; Digest of Education Statistics, table 209.10; respondent count from NCES 2024-024). Stars test the CPS mean against the published NTPS estimate with design-adjusted standard errors: *** $p < 0.001$; the female share does not differ significantly.

Full sample: **5.0 million** person-year records, 1998–2025; **104,545** teacher observations; **48,842** teachers followed across interviews in the linked panel.



The March supplement (ASEC)

- The largest supplement of the CPS: the March sample is expanded to roughly 90,000 households.
- Every adult reports on the *previous calendar year*: longest job held, weeks worked, earnings, benefits.
- We use it to reconstruct the leaving rate.

Who counts as a teacher

Occupation of the longest job held last year, CPS ASEC — three Census classifications, 28 survey years

Surveys 1998-2002

teaching years 1997-2001

1990 Census classification

Surveys 2003-2019

teaching years 2002-2018

2002/2010 Census classification (SOC)

Surveys 2020-2025

teaching years 2019-2024

2018 Census classification (SOC 2018)

155	Teachers, prekindergarten and kindergarten	2300	Preschool and kindergarten teachers	2300	Preschool and kindergarten teachers
156	Teachers, elementary school†	2310	Elementary and middle school teachers†	2310	Elementary and middle school teachers†
157	Teachers, secondary school†	2320	Secondary school teachers	2320	Secondary school teachers
158	Teachers, special education	2330	Special education teachers	2330	Special education teachers
159	Teachers, n.e.c. (not elsewhere classified)	2340	Other teachers and instructors	2360	Other teachers and instructors

*Never included (they carry their own codes, outside the set): postsecondary teachers (2200s) · tutors (2350, 2018 cl.)
teaching assistants (2540/2545) · childcare workers (4600) · education administrators (0230)*

Are the codes traceable across census years?

- Yes** as a block: the K-12 teaching group maps one-to-one across the three classifications. Every figure uses the union of the codes, never a sub-code across regimes.
- † sub-codes are not: middle-school teachers sit under elementary or secondary before 2003 and with elementary (2310) after; “other teachers” is renumbered 2340 → 2360 in 2020.
- Yes** within each survey: last year’s job (OCCUP) and the current job (PEIOCC) share one classification, so no leaver/switch comparison straddles a code change (seam surveys: 2003 and 2020).
- Yes** for people, but only two years: the CPS links the same person across two consecutive March interviews (PERIDNUM), never longer.

Two ways to measure the leaving rate

Follow the person, year to year

The rotation shows the same individual twelve months apart. The CPS has no person identifier, so records are linked on fields common to both years:

household identifier

dwelling identifier

person line number within the household

validated on sex, race,

and age advancing 0–2 years

Ask about last year, each March

The supplement's retrospective question. Whoever reports teaching as last year's longest job, and no longer teaches at the March interview, left within the year.

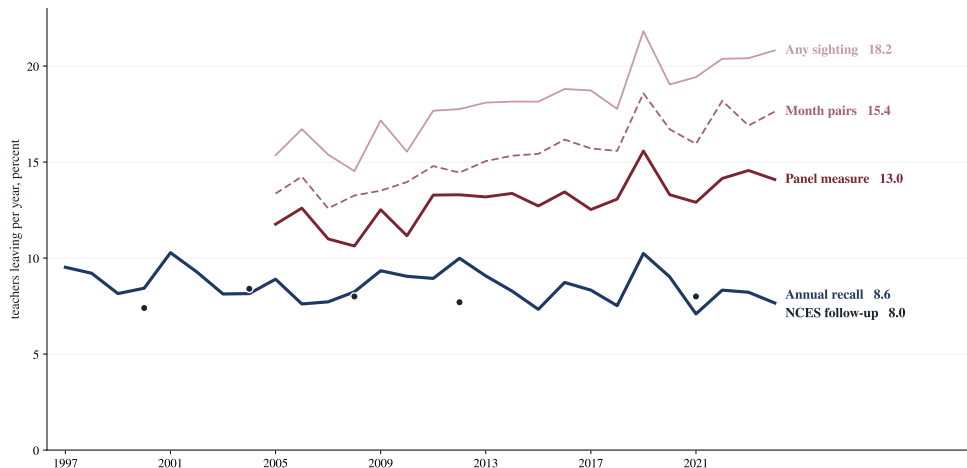
The measure used in the literature so far.

Measuring the leaving rate: the alternatives, side by side

Measure	A teacher is...	A leaver is...	Rests on...
Annual recall	whoever taught as last year's longest job	no longer teaching at the March interview	accurate recall of the previous year
Occupation pair	a teaching occupation code one year ago	a different occupation code today	a current code existing; fails for the non-employed
Panel (benchmark)	teaching in two or more year-1 interviews	never teaching in year 2, given two or more interviews, one outside the summer	correct linkage of persons across waves
Month pairs, any sighting	every teaching month, or a single sighting	not teaching twelve months after each month	one person counting several times; admits one-month spells

Same records throughout; only the definition changes. Each row answers the same three questions.

One dataset, one rate per definition



Message. The definition of a leaver moves the measured rate by a factor of nearly four.

Policy. Comparisons across studies and countries must hold the definition fixed.

Why teachers leave: what the CPS can and cannot see

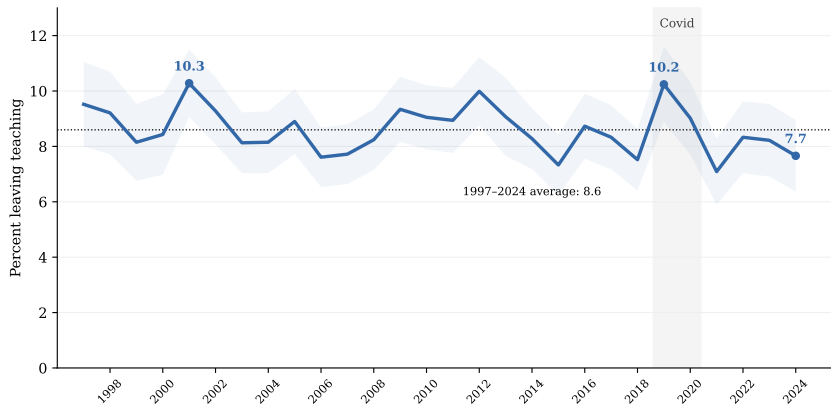
Reasons identified in the literature	Observed in the CPS?
Working conditions	no
Organizational support (mentoring, leadership)	no
Autonomy and intrinsic motives	no
Financial factors	yes
Personal life-cycle motives	yes
Sociodemographic characteristics	yes

Five broad families of reasons from our review of the literature. The CPS covers the economic and life-cycle margins.

Five messages

1. Economic and employment conditions correlate with leaving far more than demographics.
2. Exits concentrate where attachment and annual pay are lowest: part-time and part-year teachers in the bottom of the earnings distribution.
3. Those same exits lead *out of employment*, not to better-paid jobs: two thirds of bottom-quintile leavers exit the labor force. Low pay and non-employment destinations are one phenomenon, not two.
4. Life-cycle events are the largest single trigger: childbirth raises a woman's exit probability by about six points, more than in comparable professions.
5. Pension coverage retains, and attrition differs widely across states: the actionable margins are contractual and institutional.

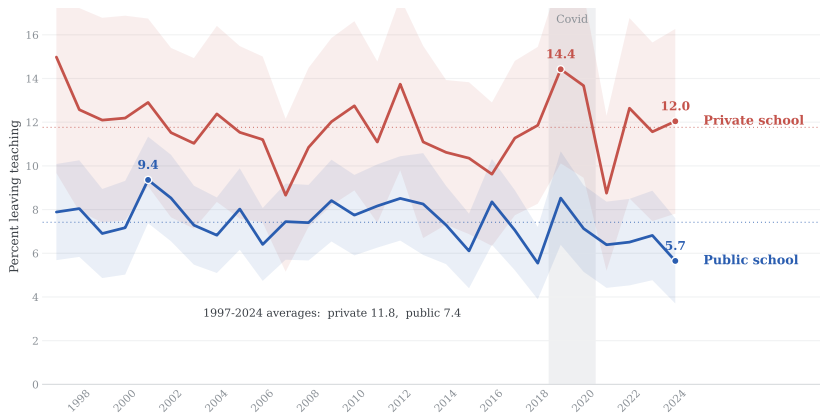
Leaving teaching, 1997–2024



Message. The rate fluctuates around 8.6 percent with no trend: a persistent outflow, not a mounting exodus.

Policy. Retention faces a steady-state cost, not an emergency spike.

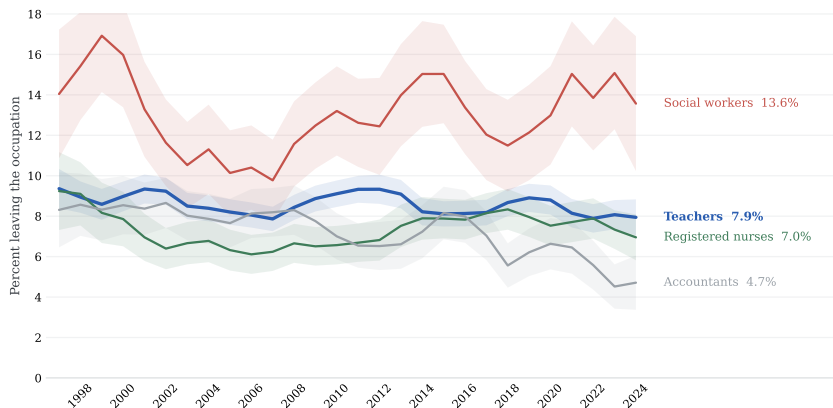
Leaving teaching, by school sector



Message. Private schools lose teachers at roughly twice the public rate.

Policy. Retention tracks contract structure, pay and pensions, not classroom conditions alone.

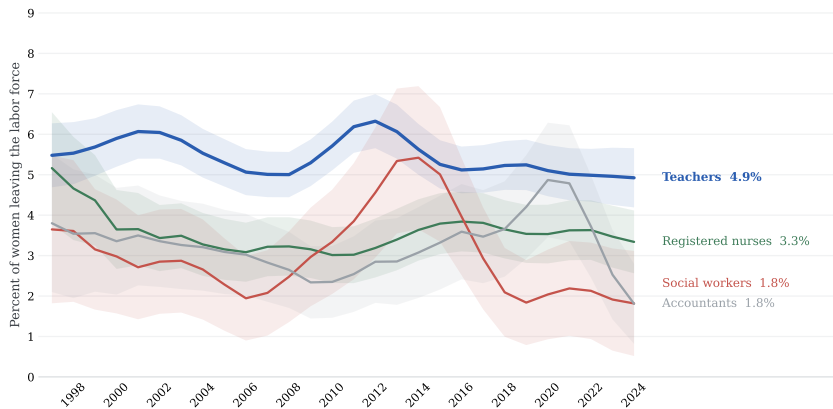
Leaving the occupation, by profession



Message. Teacher attrition is unexceptional next to comparable professions: above nurses and accountants, well below social workers.

Policy. Benchmarks from other professions discipline how alarming any teacher figure should be.

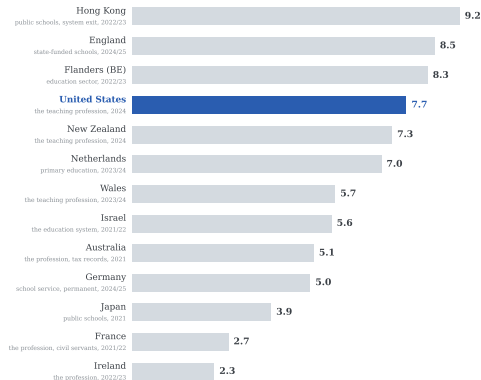
Labor force exit of women, by profession



Message. Women teachers exit the labor force at about five percent a year, well above nurses, social workers and accountants.

Policy. The family-compatibility margin is specific to teaching, not to professional work in general.

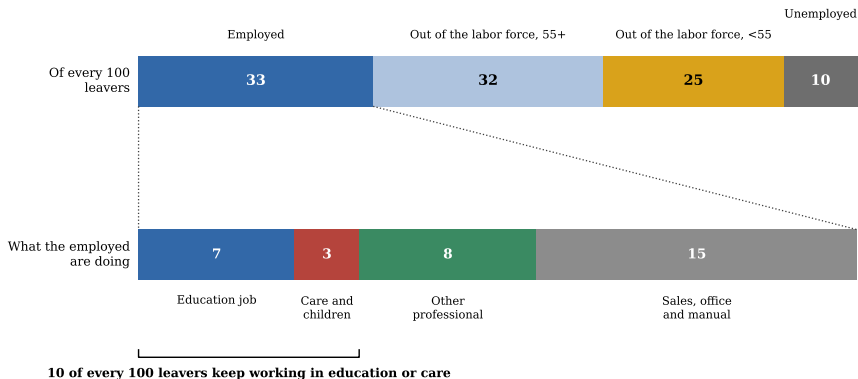
Teachers leaving per year, across countries



Message. The US sits in the upper range of the school systems that publish a rate.

Policy. Comparable, survey-based measurement is what makes this ranking possible.

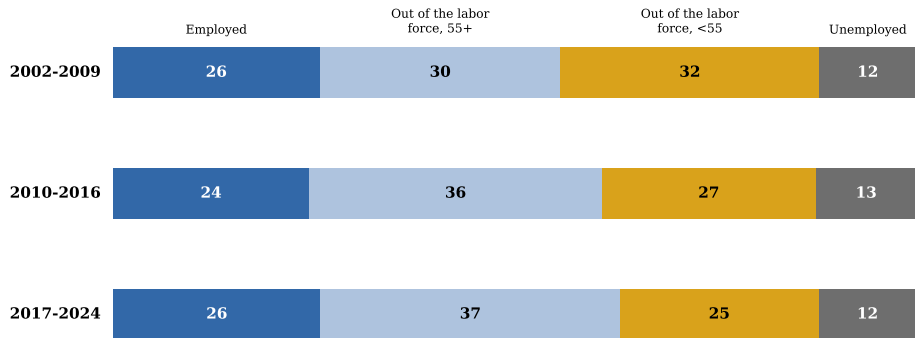
Destinations of teachers who leave, 2024



Message. Of every hundred leavers, about a third take another job; most exit employment altogether.

Policy. Competing employers are not the main counterfactual; retirement and care are.

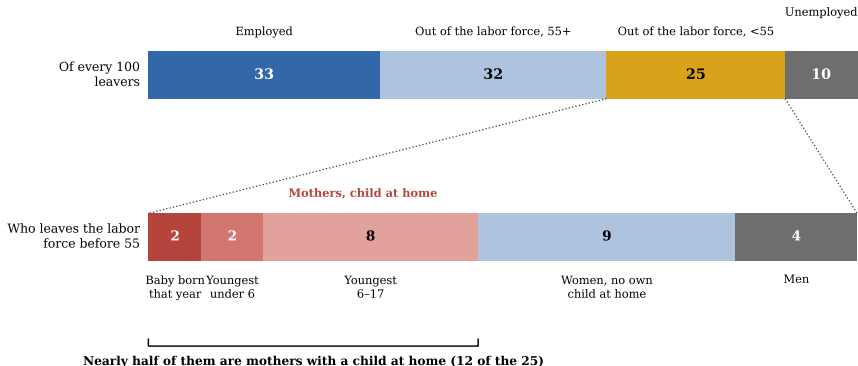
The routes out of teaching, 2002–2024



Message. The destination mix has been essentially stable for two decades.

Policy. The composition of the outflow is structural, not cyclical noise.

Who leaves the labor force before 55



Message. Nearly half of the pre-55 labor-force exits are mothers with a child at home.

Policy. Childcare and parental-leave policy are retention policy.

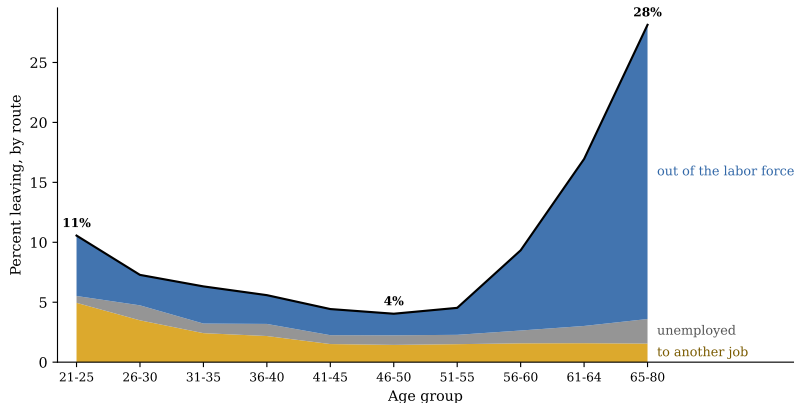
Where the employed leavers go: every occupation

EDUCATION, NON-TEACHING — 23%		PROFESSIONAL & MANAGERIAL — 32%		SALES, OFFICE & MANUAL — 36%	
8.1 Postsecondary teachers	9.7 Managers, all other	0.4 Chief executives	0.0 Miscellaneous social scientists...	0.6 Legal support workers, all other	0.3 Billing and posting clerks
4.3 Tutors	1.9 Coaches and scouts	0.4 Dietitians and nutritionists		0.6 Other healthcare support workers	0.3 Occupational health and safety S...
4.2 Teaching assistants	1.7 Architects	0.4 Information security analysts		0.6 Laborers and freight, stock, and...	0.3 Packaging and filling machine op...
3.1 Education administrators	1.5 General and operations managers	0.3 Claims adjusters, appraisers, ex...		0.6 Animal caretakers	0.2 Laundry and dry-cleaning workers
1.5 Librarians	1.3 Business operations specialists...	0.3 Physical therapists	2.8 Secretaries and admin. assistants	0.5 Insurance claims and policy proc...	0.2 Physical therapist assistants an...
1.3 Other instruction and library wo...	1.2 Other mathematical science occup...	0.3 Food service managers	1.8 Retail salespersons	0.5 Personal care aides	0.2 Janitors and building cleaners
0.2 Archivists, curators, and museum...	1.1 Writers and authors	0.3 School psychologists	1.8 Landscaping and groundskeeping w...	0.5 Bill and account collectors	0.2 Payroll and timekeeping clerks
	0.9 Broadcast, sound, and lighting t...	0.3 Editors	1.6 Exercise trainers	0.4 Waiters and waitresses	0.2 Nursing assistants
	0.8 Chemical technicians	0.3 Transportation, storage, and dis...	1.4 Supervisors of personal care and...	0.4 Data entry keyers	0.2 Helpers—production workers
	0.7 Social and community service man...	0.2 Meeting, convention, and event p...	1.3 Taxi drivers	0.4 Hairdressers, hairstylists, and...	0.1 Hosts and hostesses, restaurant...
	0.6 Property, real estate, and commu...	0.2 Training and development managers	1.3 Police officers	0.4 Real estate brokers and sales ag...	0.1 Sales representatives, wholesale...
1.8 Educational and career counselors	0.6 Photographers	0.2 Accountants and auditors	1.2 Executive secretaries	0.4 Customer service representatives	0.1 Security guards and gambling sur...
1.2 Social and human service assista...	0.6 Actors	0.2 Other healthcare practitioners a...	1.1 Stockers and order fillers	0.4 First-Line supervisors of office...	0.1 Chefs and head cooks
1.1 Mental health and substance abus...	0.6 Therapists, all other	0.2 Graphic designers	1.0 Other life, physical, and social...	0.4 Manicurists and pedicurists	0.1 Receptionists and information cl...
1.1 Social workers, all other	0.5 Lodging managers	0.2 Training and development special...	1.0 First-Line supervisors of non-re...	0.4 Other entertainment attendants a...	0.1 Computer, automated teller, and...
0.8 Childcare workers	0.5 Construction managers	0.2 Materials engineers	0.9 Title examiners, abstractors, an...	0.4 Construction laborers	0.1 Maintenance and repair workers...
0.6 Mental health counselors	0.5 Miscellaneous health technologi...	0.2 Civil engineers	0.8 Personal care and service worker...	0.4 Office and administrative suppor...	0.1 Office clerks, general
0.5 Counselors, all other	0.4 Other psychologists	0.1 Other designers	0.7 Fast food and counter workers	0.3 Cooks	0.0 Stationary engineers and boiler...
0.5 Substance abuse and behavioral d...	0.4 Other engineering technologists...	0.1 Wholesale and retail buyers, exc...	0.7 First-Line supervisors of retail...	0.3 Bookkeeping, accounting, and aud...	0.0 Tellers
0.4 Other community and social servi...	0.4 Management analysts	0.1 Physical scientists, all other	0.7 Couriers and messengers	0.3 Miscellaneous agricultural worke...	0.0 Ushers, lobby attendants, and ti...
0.4 Probation officers and correctio...	0.4 Market research analysts and mar...	0.1 Project management specialists	0.7 Statistical assistants	0.3 Travel agents	
	0.4 Human resources workers	0.1 Dietetic technicians and ophthal...	0.7 Paralegals and legal assistants	0.3 Sewing machine operators	
	0.4 Veterinarians	0.0 Floral designers	0.6 Cashiers	0.3 Bartenders	
		0.0 Farmers, ranchers, and other agr...	0.6 Recreation workers	0.3 Conveyor, dredge, and hoist and...	
		0.0 Landscape architects	0.6 First-line supervisors of landsc...	0.3 Boilermakers	

Message. Employed leavers scatter widely; the largest block stays in education outside the classroom.

Policy. Skills remain in the education orbit, so re-entry is a realistic margin.

Leaving teaching, by age and route



Message. Leaving is U-shaped in age, and non-employment dominates both ends of the career.

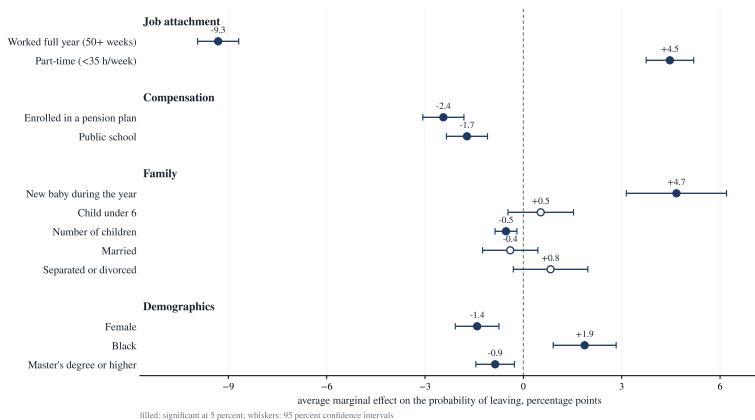
Policy. Early-career and pre-retirement exits call for different instruments.

The econometric specification

$$\Pr(\text{Leave}_{it} = 1 \mid x_{it}) = \Phi\left(\underbrace{\text{attach}'_{it}\beta_1}_{\substack{\text{full year,} \\ \text{part-time}}} + \underbrace{\text{comp}'_{it}\beta_2}_{\substack{\text{pension,} \\ \text{public school}}} + \underbrace{\text{fam}'_{it}\beta_3}_{\substack{\text{new baby, children,} \\ \text{marital status}}} + \underbrace{\text{dem}'_{it}\beta_4}_{\substack{\text{age, age}^2, \text{sex,} \\ \text{race, education}}} + \delta_t\right)$$

- Probit on college-graduate teachers, pooled 2015–2024, with calendar-year effects δ_t ; $n = 28,900$.
- Average marginal effects reported, in percentage points.
- Robust to logit and weighted LPM, state and year fixed effects, the full 1997–2024 window ($n = 85,430$), and the occupation-pair definition of leaving.

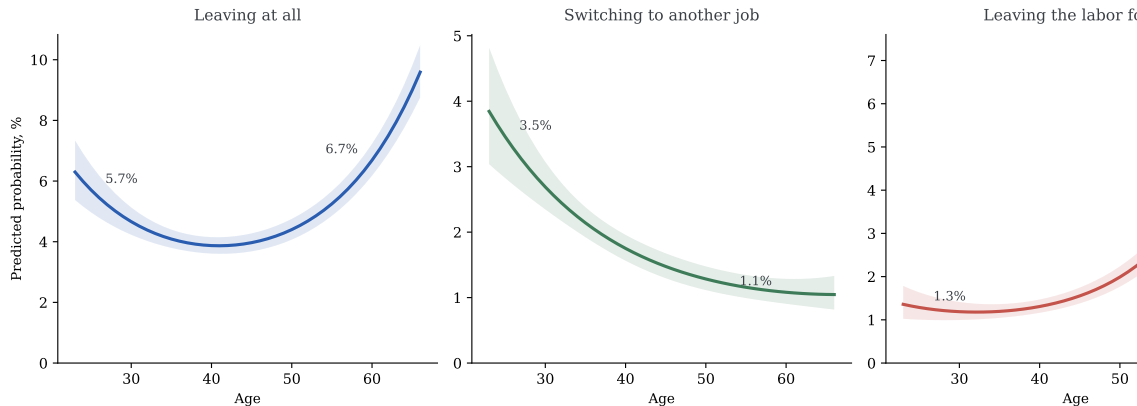
Average marginal effects on the probability of leaving



Message. Job attachment and compensation dominate; demographic differences are small.

Policy. The actionable levers are contractual: full-year contracts, pension coverage.

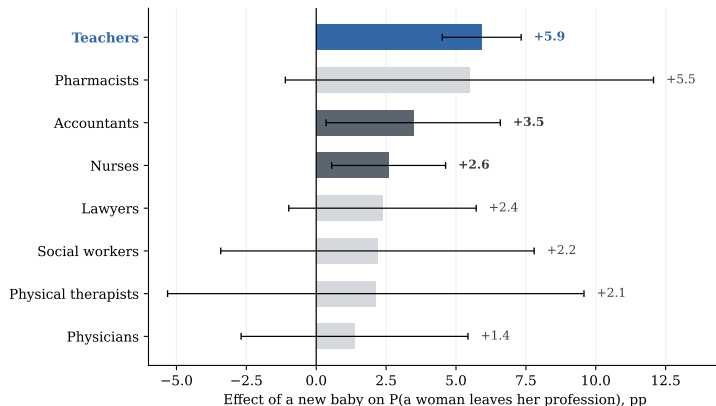
The age profile of leaving, by route



Message. Young teachers switch to other jobs; older teachers leave the labor force.

Policy. The two peaks of the U respond to different policies.

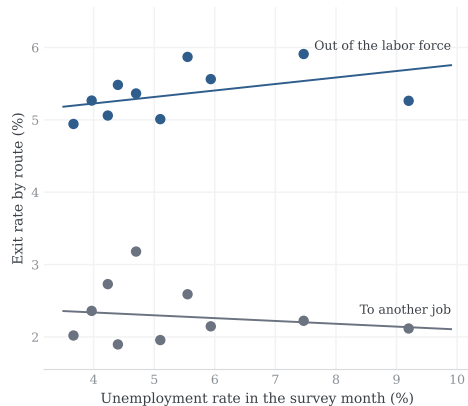
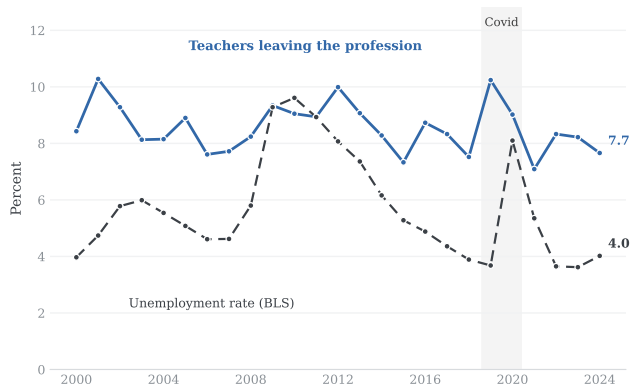
The effect of a new baby, by profession



Message. A new baby raises a woman teacher's exit probability by 5.9 points, the largest effect among comparable professions.

Policy. Parental leave and childcare support target the largest single observable trigger.

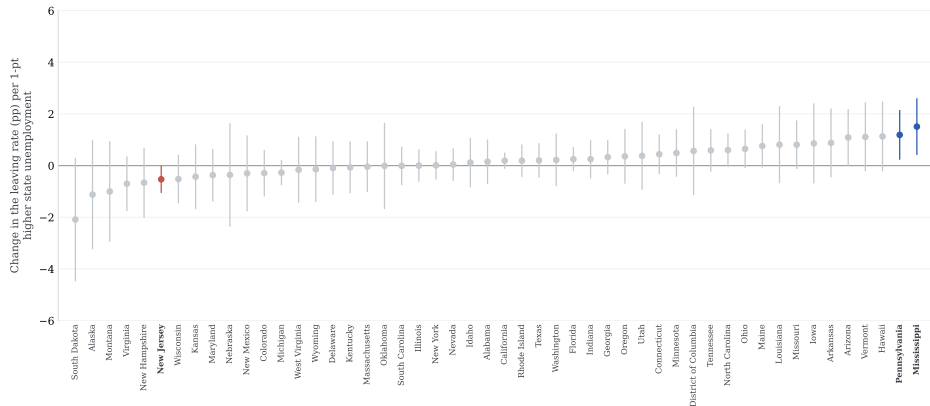
Leaving teaching and the unemployment rate



Message. Exits into non-employment rise with unemployment; job-to-job exits do not.

Policy. Downturns intensify the non-employment route; support should be countercyclical.

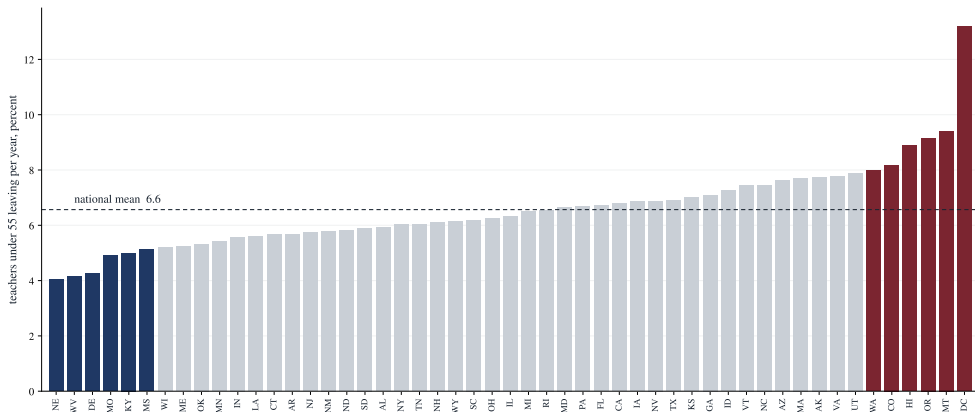
State-level cyclical sensitivity of leaving



Message. States differ in how strongly teacher exits respond to the local cycle.

Policy. National averages hide state-level exposure to downturns.

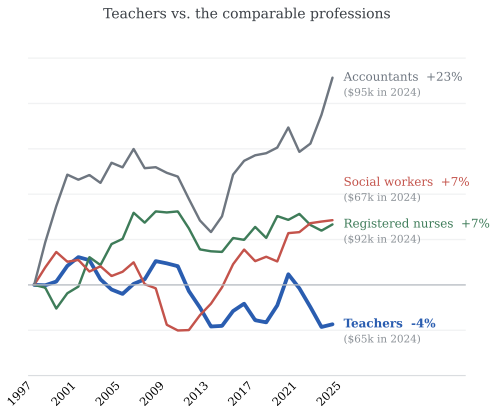
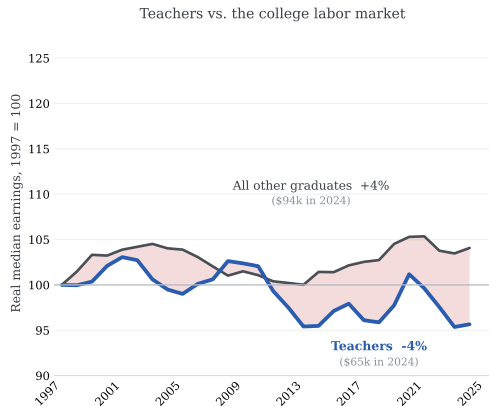
Attrition levels across states



Message. Mean attrition of teachers under 55 ranges from about 4 to 13 percent across jurisdictions.

Policy. State pay scales, pensions and institutions are first-order.

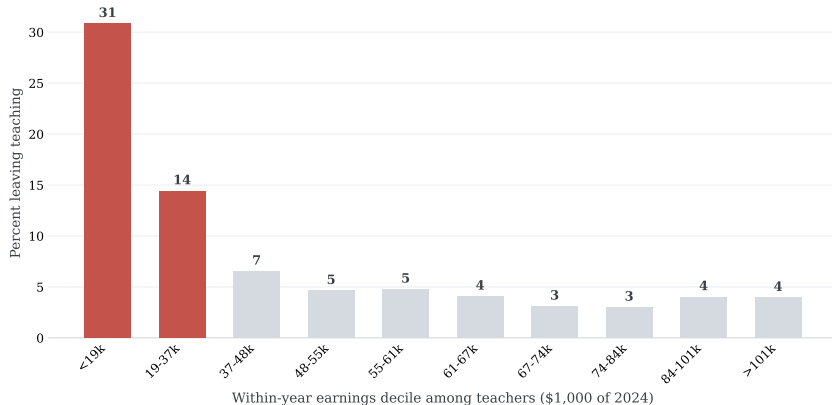
Real earnings growth by profession since 1997



Message. Teachers' real median earnings fell 4 percent since 1997 while other graduates gained.

Policy. The relative-pay gap compounds slowly; it is a stock, not a shock.

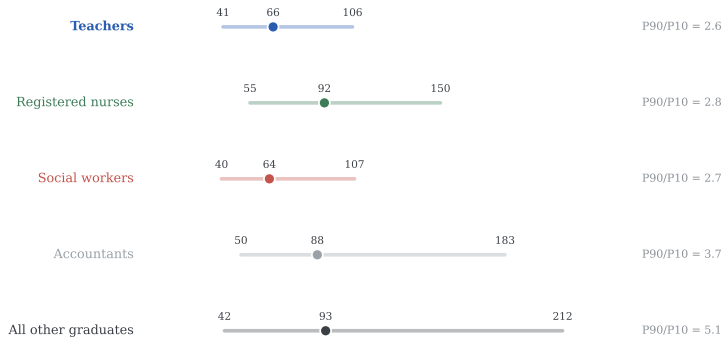
Leaving, by position in the teacher pay distribution



Message. Leaving concentrates sharply in the lowest earnings deciles, mostly part-year teachers.

Policy. Stabilizing the bottom of the distribution buys the most retention.

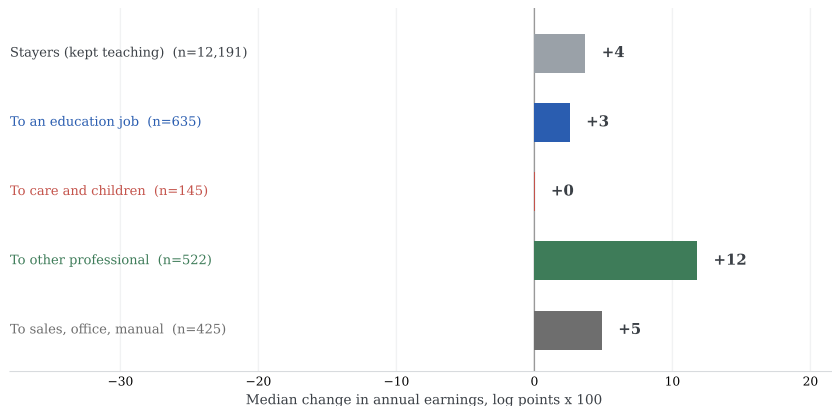
Pay compression: teachers and comparable professions



Message. Teacher pay is compressed: a 90/10 ratio of 2.6 against 5.1 for all other graduates.

Policy. Staying offers little earnings upside; career ladders would widen the right tail.

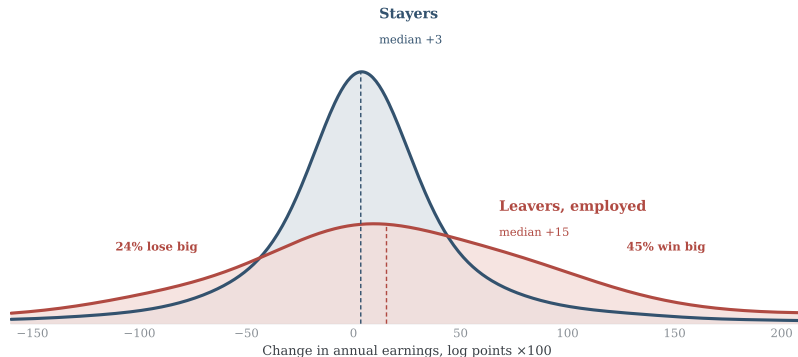
The earnings change of leaving, by destination



Message. Leavers who move to other professional jobs gain about 12 log points; moves near education gain little.

Policy. The outside option is real for those able to take it.

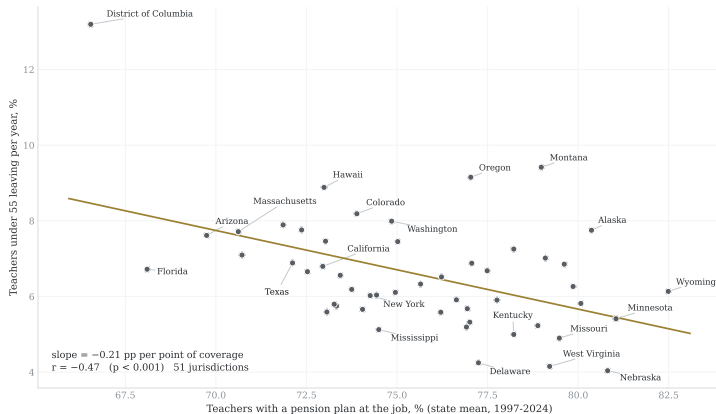
Earnings changes: stayers and leavers



Message. For employed leavers the earnings change is a gamble: a quarter lose big, nearly half win big.

Policy. Leaving buys risk, not a sure raise.

Pension coverage and leaving, across states



Message. Where pension coverage is higher, fewer teachers under 55 leave: 0.21 points per point of coverage.

Policy. Pension design is a measurable retention margin.

Conclusion

- A labour force survey can measure teacher attrition at the person level, yearly, over three decades, and it replicates the official picture of the workforce.
- The definition of a leaver is a first-order choice: the same records yield rates from 5 to 19 percent per year.
- What the data say: attachment, pay and pensions, and life-cycle events correlate with leaving; demographics much less.
- The design travels: the same construction is feasible in the labour force surveys of Italy, Germany, the UK, France, Spain, Brazil and Mexico.